Le Nguyen Gia Hung

AI/ML Researcher

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ORCID · Scholar · Scopus · Web of Science

Education

FPT University *Ho Chi Minh City, Vietnam*
Bachelor of Engineering in Artificial Intelligence *Expected Dec 2027*
GPA: 8.0/10.0

- **Honors:** Certificates of Merit – Honorable Student of Trimester (Semesters 3, 4 & 5).
- **Relevant Coursework:** Data Structures and Algorithms, Linear Algebra, Introduction to Artificial Intelligence, Deep Learning, Statistical Analysis.

Publications

Gia-Hung Nguyen Le *et al.*, "Proactive Air Quality Forecasting and Health Alert System for Melbourne." *Australasian Joint Conference on Artificial Intelligence (AJCAI 2025)*. Springer, Lecture Notes in Artificial Intelligence (LNAI). *Q2 (SJR 2024)*.
DOI: 10.1007/978-981-95-4969-6_34

Research & Technical Experience

Proactive Air Quality Forecasting System (AJCAI 2025) *Academic Research*
Lead Researcher & Author *Jan 2025 – Present*

- Architected and implemented **SmokeNet**, a novel Transformer-based architecture for time-series forecasting of $PM_{2.5}$.
- Achieved **MAE of 0.7470** and R^2 **of 0.9545**, reducing error by **35.9%** compared to XGBoost baselines on 2024 test data.
- Demonstrated model robustness by outperforming baselines by **57.7%** during the severe May 2024 smoke event backtest.
- Engineered an end-to-end pipeline processing 4+ years of OpenWeather data with GPU-accelerated chronological cross-validation.
- Integrated Explainable AI (SHAP, Integrated Gradients) to provide interpretable health warnings for public safety dashboards.

MERR-GAT: Explainable Multimodal Emotion Recognition *Academic Research*
Corresponding Author & Co-Lead *In Progress*

- Developing **MERR-GAT**, a novel hybrid framework fusing **Wav2Vec 2.0** (audio) and **RoBERTa** (text) via Graph Attention Networks (GATv2).
- Designing a dynamic conversational graph where nodes represent utterances and edges capture temporal and speaker interactions.
- Implementing an intrinsic XAI mechanism where learned attention weights provide granular insights into the model's decision-making.
- Benchmarking on **IEMOCAP**, **RAVDESS**, and **MELD** with a dynamic classification head supporting 4, 5, or 6 emotion classes.

End-to-End Vietnamese Speech Recognition System *Academic Research*
Co-Researcher *In Progress*

- Developing a parameter-efficient (5.4M) end-to-end ASR model using a **4-layer CNN frontend** and **3-layer Residual BiLSTM encoder**.
- Curated and trained on a massive **745-hour Vietnamese speech corpus**, optimizing with Connectionist Temporal Classification (CTC) loss.

- Achieved a **Word Error Rate (WER) of 33.09%** and **Character Error Rate (CER) of 15.28%** on diverse test sets.
- Benchmarked performance against foundation models (PhoWhisper) to evaluate trade-offs between computational efficiency and accuracy.

Personal Protective Equipment (PPE) Detection System *Course Project*
Developer *Sep 2024 – Dec 2024*

- Built a real-time object detection system using **YOLOv8** to identify 9 classes of personal protective equipment (helmets, vests, gloves).
- Developed a safety logic engine to automatically classify workers as "Compliance Safe" or "Unsafe" based on detected equipment combinations.
- Deployed the model via a **Streamlit** web interface featuring live webcam inference and automated violation reporting.

Leadership

SpeedyLabX - Student Research Group *FPT University*
Founder & Lead Researcher *2025 – Present*

- Established and directed a research team of 10 undergraduates focused on applied AI solutions for environmental challenges.
- Spearheaded the Melbourne Air Quality project, managing timeline and deliverables, resulting in the group's first accepted publication at a top-tier regional conference (AJCAI).

Certifications

- **Building RAG Agents with LLMs –** NVIDIA
- **IBM Full Stack Software Developer Professional Certificate –** IBM
- **Data Science Fundamentals with Python and SQL –** IBM

Skills

- **Programming Languages:** Python, SQL (PostgreSQL), C++.
- **Frameworks & Libraries:** PyTorch, TensorFlow, Keras, Hugging Face Transformers, Scikit-learn, OpenCV.
- **Data & Tools:** Pandas, NumPy, Matplotlib, Git, Docker, Streamlit, Weights & Biases (W&B), Linux.
- **Technical Writing:** Authored "My First Encounter with AI" (Personal Tech Blog).
- **Areas of Interest:** Time-Series Forecasting, Transformer Architectures, Multimodal Learning, Explainable AI (XAI).